HEART RATE MONITOR

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HEART RATE MONITORS

- A heart rate monitor (HRM) is a personal monitoring device that allows one to measure/display heart rate in real time or record the heart rate for later study.
- It is largely used to gather heart rate data while performing various types of physical exercise.
- Measuring electrical heart information is referred to as Electrocardiography (ECG or EKG).

- Medical heart rate monitoring used in hospitals is usually wired and usually multiple sensors are used.
- Portable medical units are referred to as a Holter monitor.
- Consumer heart rate monitors are designed for everyday use and do not use wires to connect.

METHODS

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- Modern heart rate monitors commonly use one of two different methods to record heart signals (electrical and optical).
- ECG (Electrocardiography) sensors measure the bio-potential generated by electrical signals that control the expansion and contraction of heart chambers, typically implemented in medical devices.
- PPG (Photoplethysmography) sensors use a light-based technology to measure the blood volume controlled by the heart's pumping action.

ELECTRICAL

- The electrical monitors consist of two elements: a monitor/transmitter, which is worn on a chest strap, and a receiver.
- When a heartbeat is detected a radio signal is transmitted, which the receiver uses to display/determine the current heart rate.
- This signal can be a simple radio pulse or a unique coded signal from the chest strap.

OPTICAL

- More recent devices use optics to measure heart rate by shining light from an LED through the skin and measuring how it scatters off blood vessels.
- In addition to measuring the heart rate, some devices using this technology are able to measure blood oxygen saturation (SpO_2).
- Newer devices such as cell phones or watches can be used to display and/or collect the information.
- Some devices can simultaneously monitor heart rate, oxygen saturation, and other parameters.
- These may include sensors such as accelerometers, gyroscopes, and GPS to detect speed, location and distance.

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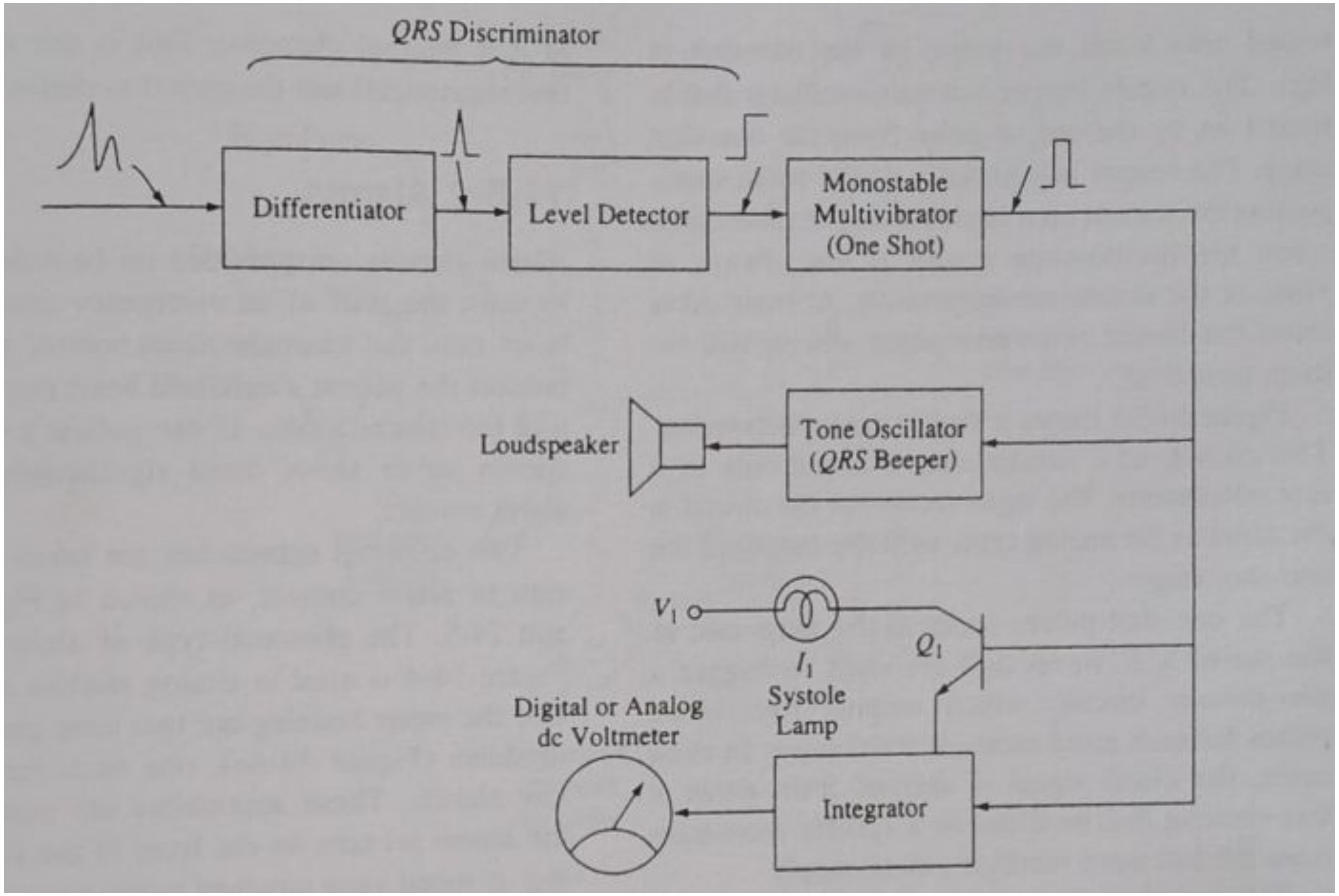
HEART RATE MONITORS

CARDIOTACHOMETERS

- It is a heart rate meter.
- Provides an analog or digital display of the heart rate.
- Precisely relies on ECG data
- Used in Hospitals

ANALOG TYPE



QRS Discriminator
Differentiator
Level Detector
Monostable
Multivibrator
(One Shot)
Loudspeaker
Tone Oscillator
(QRS Beeper)
V_1
I_1
Systole
Lamp
Q_1
Digital or Analog
dc Voltmeter
Integrator

ANALOG TYPE

- **ECG Input**

 The raw ECG signal contains P wave, QRS complex, and T wave.

 The QRS complex has the **highest slope and amplitude**, making it ideal for detection.

- **Differentiator**

 Emphasizes **rapid changes** in the ECG signal.

Since the QRS complex has a steep slope, it produces **large spikes**, while P and T waves are suppressed.

Output: sharp positive/negative pulses corresponding mainly to QRS complexes.

ANALOG TYPE

- **Level Detector**

 Acts as a **threshold detector**.

 Only signals exceeding a preset voltage level (from QRS spikes) are allowed through.

Rejects noise and smaller ECG components.

Output: digital-like pulses indicating QRS detection.

- **Monostable Multivibrator (One-Shot)**

 Converts each detected QRS pulse into a fixed-width pulse.

Ensures One output pulse per heartbeat, Prevents double triggering due to noise

This pulse becomes the master timing signal for further actions.

ANALOG TYPE

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- **Tone Oscillator (QRS Beeper)**
  - Activated by the one-shot pulse.
  - Generates an audible **“beep”** for each heartbeat.
  - Output goes to the **loudspeaker**.

- **Systole Lamp (via Transistor Q_1)**
  - The one-shot pulse switches transistor **Q_1 ON**.
  - The **systole lamp** lights up during ventricular contraction.
  - Provides a **visual indication** of each heartbeat.

ANALOG TYPE

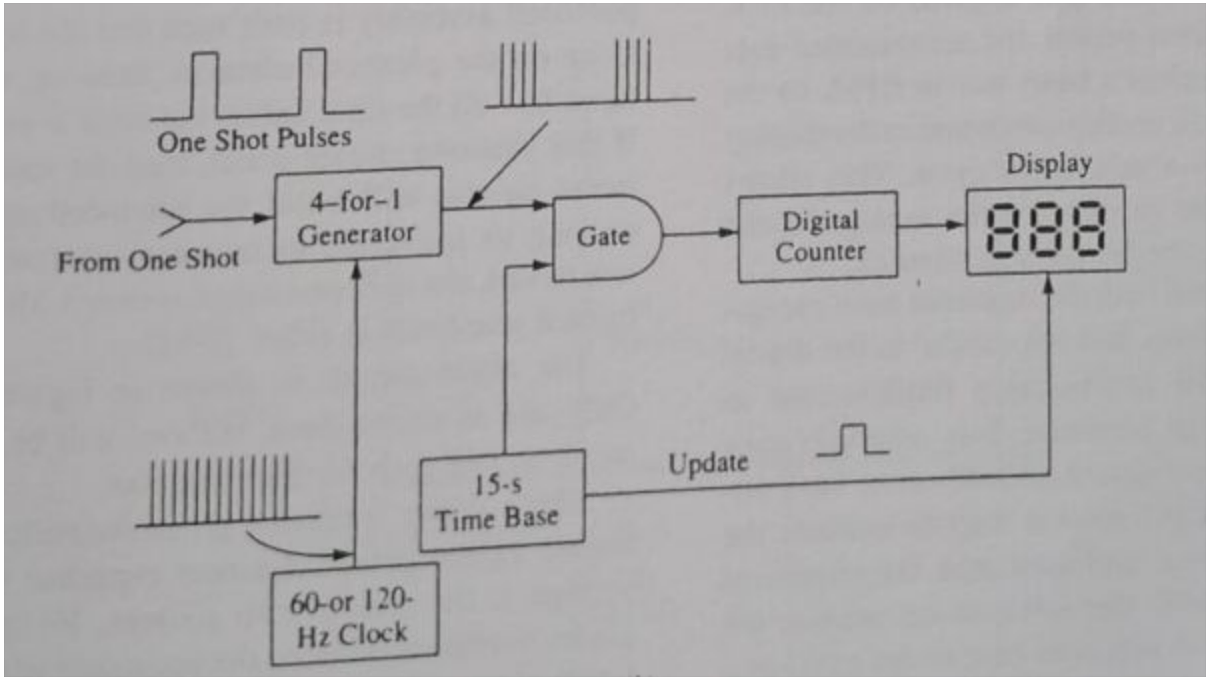
- **Integrator**

Integrates the train of pulses from the one-shot.
Output voltage is proportional to **pulse frequency**, i.e., heart rate.

- **DC Voltmeter (Digital or Analog)**

 Measures the integrator output voltage.
Calibrated directly in **beats per minute (BPM)**.
Displays the patient’s heart rate.

DIGITAL TYPE



LINK

- https://www.youtube.com/watch?v=RHxfP5rm3Ks
- https://www.youtube.com/watch?v=tq_hz8FjStA

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HOLTER MONITOR

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HOLTER MONITORS

- Continuous ECG recording.
- Normal ECG – cardiac activity under certain conditions.
- Introduced by Norman Holter – capture arrhythmias.
- Continuous ECG – recorded from patient during his normal daily activity by means of a special magnetic tape recorder.

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- Smallest device which can be worn in a shirt pocket and allows recording of the ECG for 4 hours.
- Other recorders – worn over the shoulder can record the ECG up to 24 hours.
- Recorded tape is analyzed using a special scanning device which plays back the tape at a higher speed than that used for recording.
- By this method, a 24 hour tape can be reviewed in 12 minutes.

- During the playback, the beat to beat interval of the ECG is displayed on a cathode ray tube as a picket fence like pattern in which arrhythmia are clearly visible.
- Once arrhythmia is discovered, the tape is backed up and slowed down to obtain a normal ECG strip for the time interval during which the arrhythmias occurred.
- A special time clock is synchronized by the tape drive to correlate the onset of the arrhythmias with the activity of the patient.

HOLTER MONITORS – DATA RECORDING

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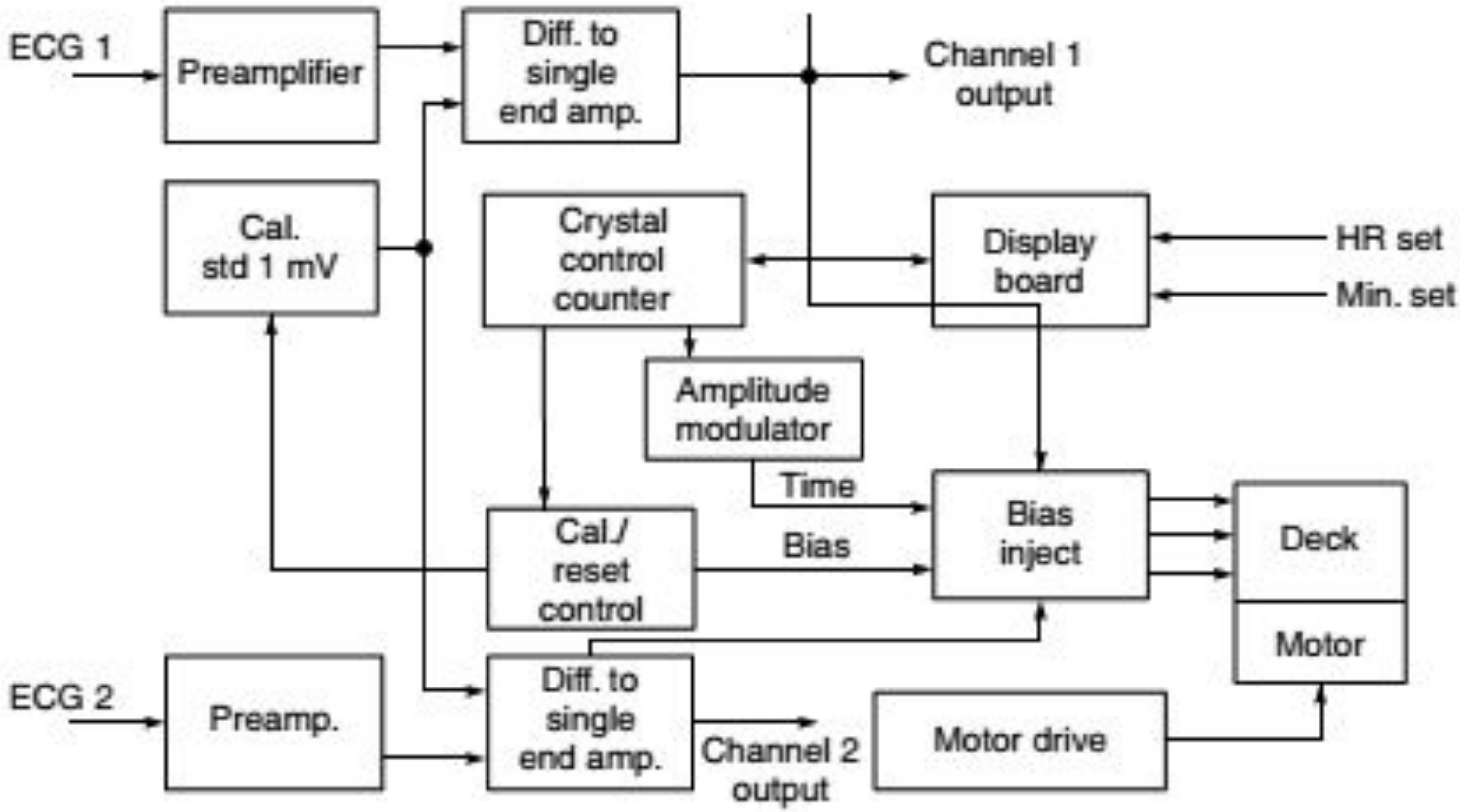
- The core of the modern ambulatory monitoring system is a multi-channel sub-miniature tape recorder running normally at a speed of 2 mm/s.
- At this speed, a C-120 entertainment cassette will run for 24 h.
- The recorders are designed to be fitted with a variety of plug-in circuit boards adapted for different signals or transducers.
- The main areas of interest in ambulatory monitoring are centred on the cardiovascular system, with particular reference to the control of cardiac rhythm disturbances, along with treatment of hypertension and the diagnosis and treatment of ischaemic heart disease.

HOLTER MONITORS – DATA RECORDING



HOLTER MONITORS – DATA RECORDING

- During replay, the tape is run at 120 mm/s (60 times the recording speed) to achieve rapid manual or automatic scanning of ambulatory records.
- Tape recorder offers single sided 24 h recording facility with three recording channels – two for ECG and one for timing, for the precise correlation of recorded events to patient activity.
- As in conventional circuits, the ECG channels feature high input impedance, differential inputs consisting of transient protection mode switching and differential amplifier stages, followed by a single ended amplifier, with provision for fast transient recovery and reset.

HOLTER MONITORS – DATA RECORDING

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- Calibration is automatic, as it switches the ECG channels to receive the 1 mV pulse output of a calibration circuit.
- The timing channel provides duty cycle coding for odd versus even minutes as well as half hour transitions.
- Timing channel information is amplitude modulated at low frequency so that the recorder user may mark significant episodes without disturbance to either the ECG or timing information.
- Time coding is a function of a crystal controlled counter, which controls biasing to the three recording channels.
- Conventional tape and solid-state systems commonly use data compression.
- This process requires digitization at least at 250 samples/s.
- Digitization through tape or solid-state systems results in the loss of the original ECG data by compression since digitization only records certain points along the waveform.

CARDIAC PACEMAKERS

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PACEMAKERS

- An electrical impulse is generated by the SA node, located in the right atrium.
- The pulse propagates down the electroconduction system, including the AV node to the ventricles.
- The system splits into left and right bundle branches in the ventricles.
- If an interruption occurs in the electroconduction system, causing a condition called **heart block**, then the heart's ability to pump blood is disrupted or stopped.

- Physicians can maintain electrical stimulation of the heart using a pacemaker.
- Pacemaker is an electrical generator that delivers needed pulse at an appropriate time.
- **Types:**
  - External – worn on the belt or placed at the bedside (400 μJ)
  - Internal – surgical implanted inside the patient (10 μJ)

EXTERNAL PACEMAKERS

- External pacemakers are temporary measure used following open-heart surgery for certain problems experienced by some myocardial infarction patients and for patients who are to evaluated as candidates for surgical implantation of a permanent model.
- Adjustable from 50 to 150 bpm and produce fixed-duration, short duty pulses (1.5 to 2 ms).
- Peak amplitude is adjustable from 100 μA to 20 mA.

INTERNAL PACEMAKERS

- Permanent pacemakers are built into molded epoxy-silicone rubber packages.
- Recent models – outer titanium shield that guards against interference from radio frequency fields.
- Device is implanted subcutaneously in either abdomen or region just below the collarbone.
- Implantable pacemakers have single fixed rate about 70 bpm, while other have dual rate models.
- Dual rate models can be programmed from outside the patient's body using a magnet or induction coil.
- Others are programmable from 30 to 150 bpm.

LEAD WIRES

- Types:
  - Endocardial leads
  - Myocardial leads
- These categories subdivided into unipolar and bipolar.
- **Endocardial leads** – inserted through an opening in a vein and then threaded through the venous system and right atrium and into the right ventricle of the heart.
- **Myocardial leads** – connected directly to the heart.
- **Bipolar leads** – both electrodes are inside a single catheter.
- Distal tip is one electrode while the second is located at a short distance behind the tip.
- These electrode are made of platinum-iridium alloy to prevent interaction with body fluids.

BATTERIES

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Learn Beyond

- Principal power source for implantable pacemakers is the **lithium iodine cell**.
- Mercury pacemaker batteries – operate as long as 4 to 5 years.
- Pacemaker manufacturers use X-rays to screen the assembly defects.
- Batteries – weak point in pacemakers.
- Some hospitals operate pacemaker surveillance clinics to spot premature battery failure.
- Pulse rate drops with decreased battery voltage.
- Researches on nuclear power source – 10 year power pack
- Uses heat generated by the natural decay of certain radionuclides to generate electricity in a thermocouple.
- At present – mercury and lithium battery cells are widely used for implantable pacemakers.

PACEMAKER CLASSIFICATION

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- Classified based on the output pulse they produce.
- **Monopolar output pulse**
- **Biphasic pulse** – high amplitude pulse of one polarity followed by lower amplitude pulse of opposite polarity.
- Four general categories
    - Asynchronous
    - Demand
    - R-wave inhibited
    - AV synchronized

- **Asynchronous pacemaker** produces pulses at a fixed rate in the 60-80 beats/min range. (Standard rate – 70 beats/min)
- **Demand pacemaker** adjusts its firing rate to the patient’s heart rate.
- It contains circuitry that senses the ECG R wave and measures the R to R interval.
- During the first quarter of this period, the pacemaker is dormant to prevent the response to the T wave feature of ECG.
- But during the last three quarters, it is in sense mode.
- If an R wave is not sensed within this period, then the pacemaker emits a pulse.
- **R-wave inhibited pacemaker** is similar to the demand type, except that it does not emit the pulse during normal heart activity.
- Triggering circuits are inhibited for a period of time after each R wave.
- The pulse is enabled, if no R wave occurs for a preset period.

- **AV synchronized pacemaker** responds to ECG P wave (contraction of atria).
- Atrial pacer circuitry contains a P Q delay circuit that stimulates the propagation time in the heart's electroconduction system.
- Once the pacer senses a P wave, and P Q delay of 120 ms has elapsed, then the pacer will fire a pulse to stimulate the ventricles into conduction.
- AV pacer has the advantage that it follow the changing heart rate demands of the body.
- Pacers synchronized to the ventricles can result only in ventricular contractions independent of the atrial heart rate set by the SA node.
- AV pacer will usually track heart rates in the 50 to 150 beats/min range but will revert to fixed rate if the atrial rate exceeds 150 beats/min.
- This prevents atrial triggering if atrial flutter or tachycardia occurs.

DEFIBRILLATORS

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DEFIBRILLATOR

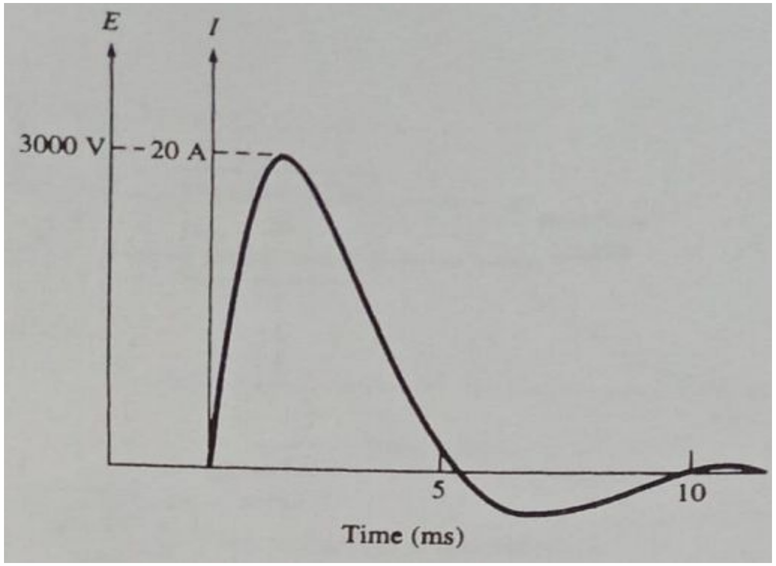
- Defibrillator – device that electric shock to the heart muscle undergoing a fatal arrhythmia.
- AC defibrillator (1960) – 5 to 6 A of 60 Hz ac for 250 to 1000 ms.
- Success rate was low.
- Attempting to correct atrial fibrillation using ac, it results in producing ventricular fibrillation.
- DC defibrillators – store a dc charge that can be delivered to the patient.

TYPES OF DC DEFIBRILLATOR

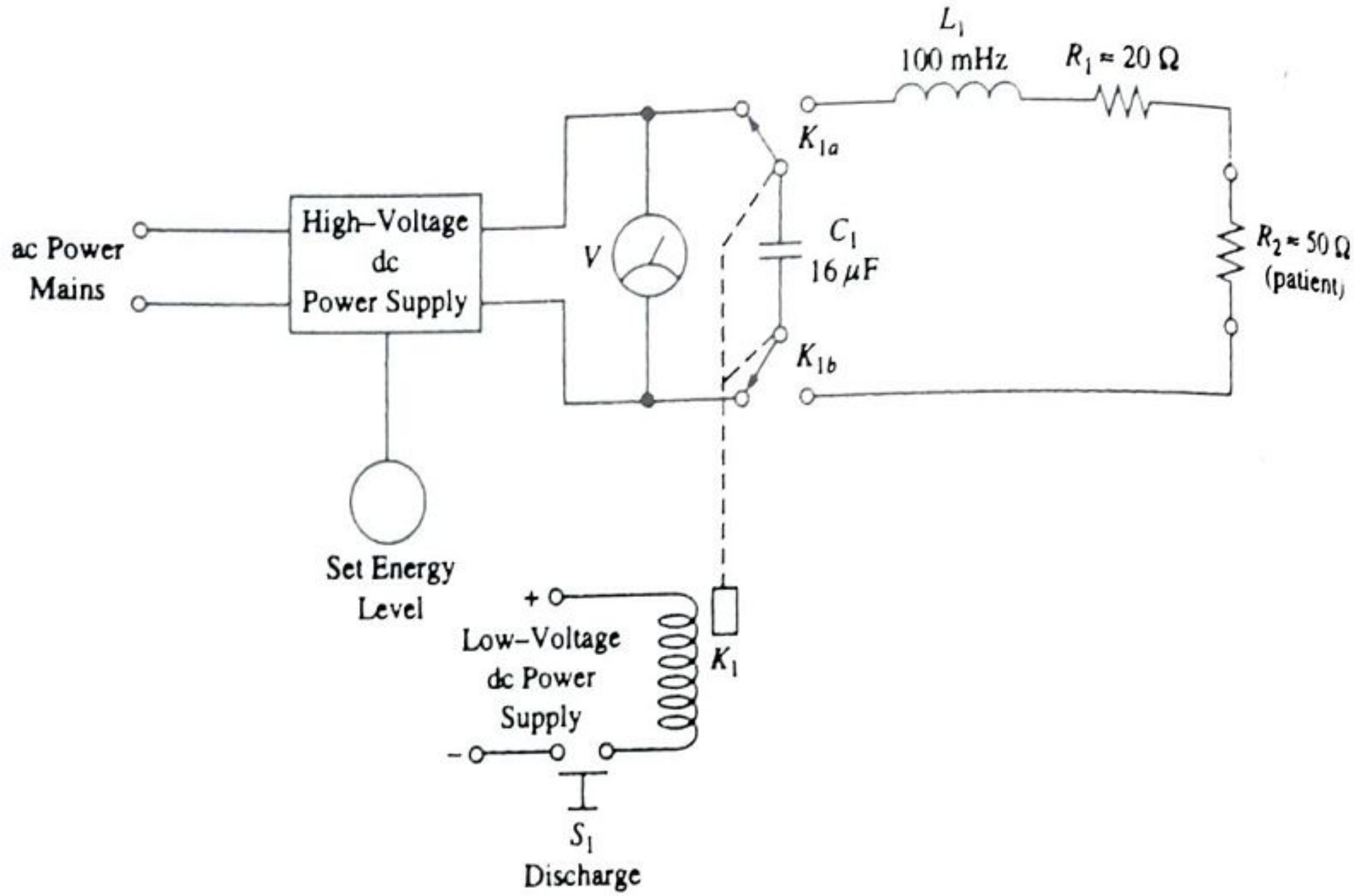
- Based on the waveshape of the charge delivered to the patient.
  - Lown
  - Monopulse
  - Tapered (dc) delay
  - Trapezoidal waveforms

LOWN DEFIBRILLATORS

- In Lown waveform, the voltage and current applied to the patient's chest is plotted against time.
- The current will raise rapidly to about 20 A under the influence of slightly less than 3 kV.
- Then the waveform decays back to zero within 5 ms, and then produces a smaller negative pulse also about 5 ms.



CIRCUIT DIAGRAM



CIRCUIT DIAGRAM

- The charge delivered to the patient is stored in a capacitor and is produced by a high voltage dc power supply.
- The operator can set the charge level using the set energy knob on the front panel.
- The knob controls the dc voltage produced by the high-voltage power supply and so can set the maximum charge on the capacitor.
- The energy stored in the capacitor is given by

$$U = \frac{1}{2} CV^2$$

where U is the energy in joules (J)
C is the capacitance of C_1 in farads (F)
V is the voltage across C_1 (V)

CIRCUIT DIAGRAM

KPRIET
Learn Beyond

- The stored energy is indicated by a voltmeter connected across the capacitor.
- The scale of the voltmeter is calibrated in energy units.
- Some energy is lost in the relay switching contacts and in the ohmic resistance of inductor L_1.
- The capacitance charge is controlled by a relay switch K_1.
- In early models, single pole-double throw (SPDT) relays were used but in all recent models double pole-double throw (DPDT) relays are used so that isolation of the patient circuit from the ground is maintained.
- Patient circuit of the Lown defibrillator consists of a 100 mH inductor (L_1), the ohmic resistance of L_1 (R_1) and the patient's ohmic resistance (R_2).
- It is the energy stored in the magnetic field of coil L1 that produces the negative excursion of the Lown waveform during the last 5 ms.
- When the capacitor has discharged, the coil's field collapses, dumping energy back into the circuit.

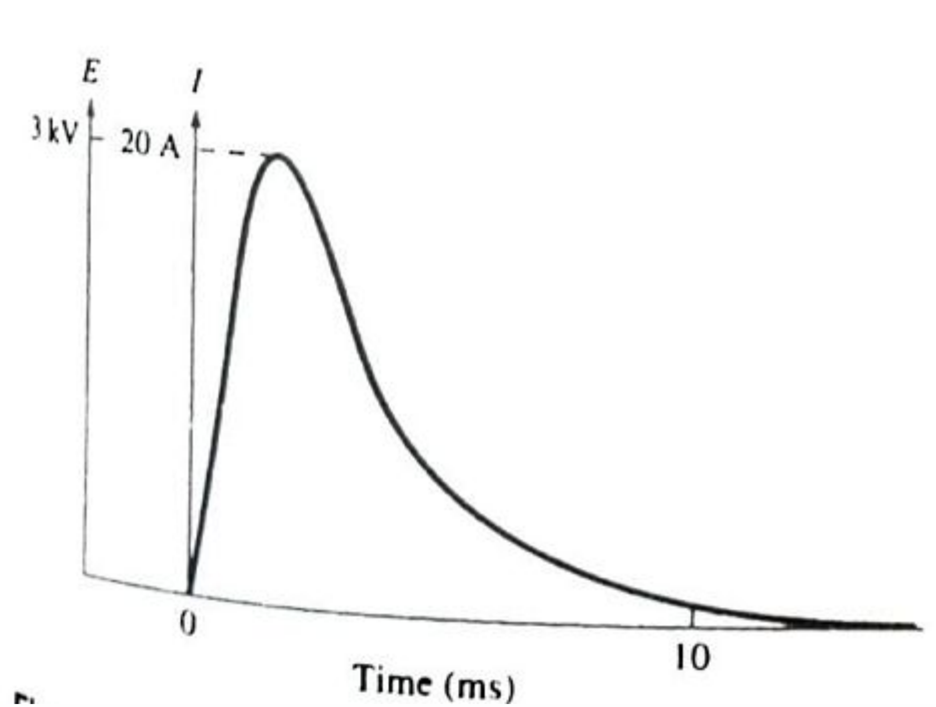
SEQUENCE OF EVENTS

- The operator turns the set energy control to the desired level and presses the charge button.
- Capacitance C_1 begins charging and will continue to charge until the voltage across the capacitor is equal to the supply voltage.
- The operator positions paddle electrodes on the patient's chest and presses the discharge button.
- Relay K1 disconnects the capacitor from the power supply and then connects it to the output circuit.
- Capacitor C1 discharges its energy into the patient through L1, R1 and the paddle electrodes.
- The magnetic field built up around L1 collapses during the last 5 ms of the waveform producing the negative excursion.

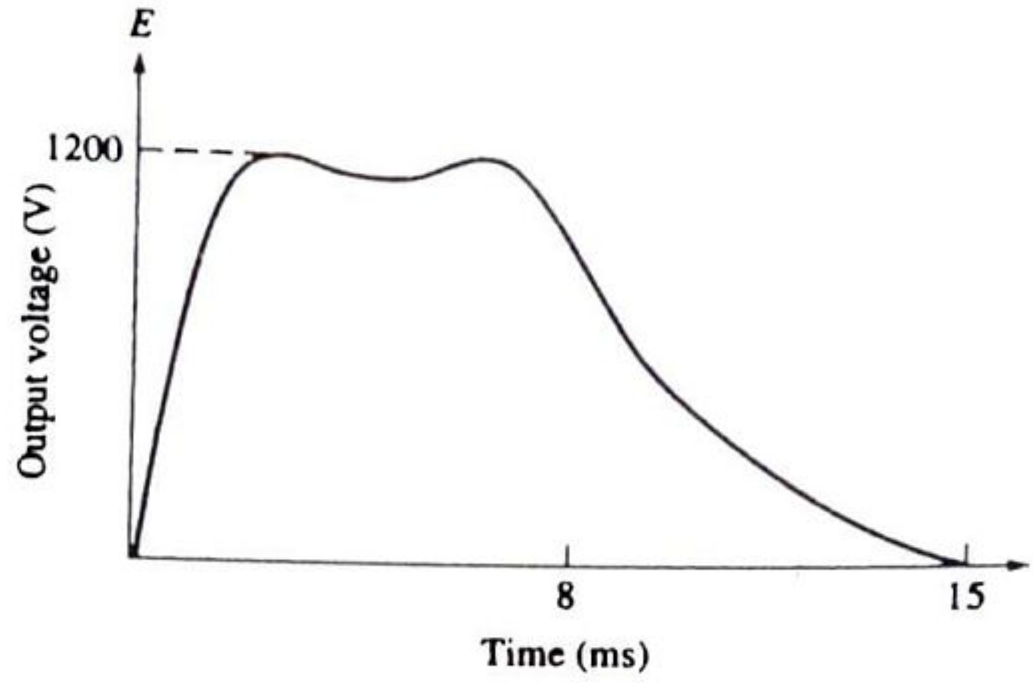
PROBLEM

- Calculate the energy stored in a 16 μF capacitor that is charged to a potential of 5000 V dc.

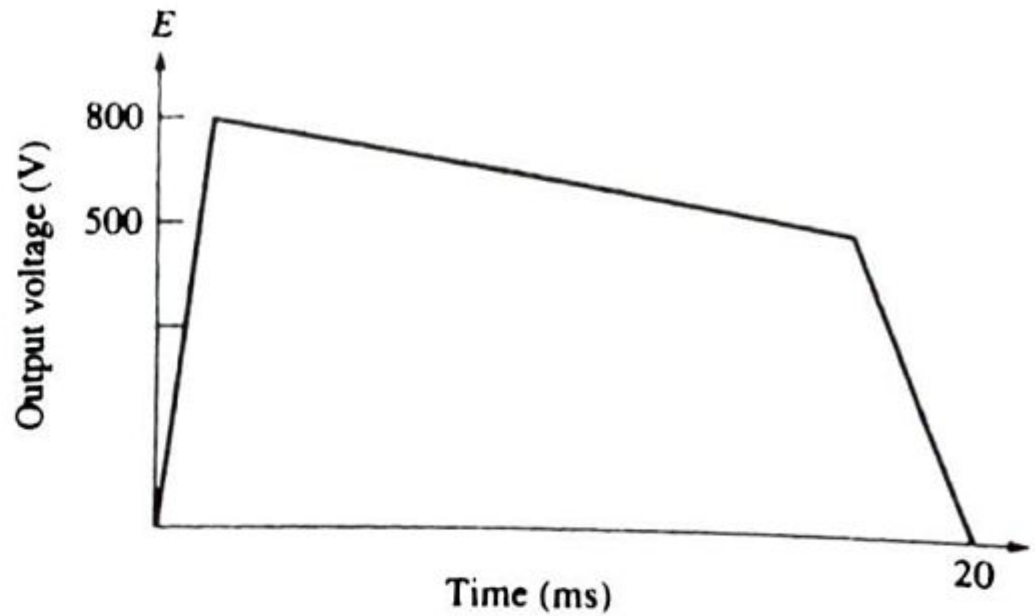
MONOPULSE DEFIBRILLATOR WAVEFORM



TAPERED DC DELAY DEFIBRILLATOR WAVEFORM



TRAPEZOIDAL DEFIBRILLATOR WAVEFORM



Example 13-7

Given the preamplifiers shown in Figure 13-15*b*, which is also ac-coupled to reduce slow baseline drift, calculate the *low-cutoff frequency (half power* or −3 dB down frequency).

Solution

$$f_{low} = \frac{1}{2\pi(R_1 \| R_{in1})(C_1)}$$

$$= \frac{1}{2\pi(5\ \text{M}\Omega)(0.1\ \mu\text{F})}$$

$$= \mathbf{0.318\ Hz} \qquad (13\text{-}9)$$

The *single-ended input impedance* in Figure 13-15*b* is approximately 5 MΩ.

EEG preamplifiers are the predominant stage that influences EEG machine specifications.

EEG machine specifications typically include:

1. Input impedance: 12 MΩ minimum at 10 Hz.
2. Sensitivity: 0.5 μV/mm maximum.
3. Sensitivity controls: 10-position master (2 to 75 μV/mm), six-position individual channel (X 20 to X 0.25), and individual-channel gain equalizer.
4. Calibration voltages: 5 to 1000 μV.
5. CMRR: 2000 or 66 dB minimum at 60 Hz and 10,000 or 80 dB minimum at 10 Hz.
6. Noise: 1 μV_{rms} (equivalent referred to input) with input shorted.
7. Low frequency (time constant): 30% attenuation—0.16 through 5.3 Hz, at time constants of one through 0.03 s, respectively.
8. High-frequency response: 30% attenuation at 1 to 1000 Hz.
9. 60-Hz filter: 50 dB down at 60 Hz.
10. Chart speeds: 10 to 60 mm/s.

13-15 Visual and auditory evoked potential recordings

Early EEG investigators discovered that *cortical potentials* could be evoked from sensory stimuli. EEG changes were noted when loud sounds were present. Although wave and spike resources, as well as K-complexes, are evident in the primary record (real time), most evoked responses on the scalp are too small to be recorded by typical EEG machines.

The technique of *averaging repetitive EEG signals* allows the tiny evoked potential to be separated from the background EEG. Each EEG response (5 μV_{p-p}) to a sensory stimulus is added to produce a readable signal. Along with this evoked signal enhancement, the ongoing EEG signal (100 μV_{p-p}) is reduced (averaged out due to its random nature). Some information is lost in this process; therefore, the number of stimulations should be kept to a minimum. Unfortunately, each response is unique, and the result can only represent the addition of all 100 different potentials.

The *block diagram* in Figure 13-16 shows a *visual and auditory evoked potential system.* Three electrodes pick up a one-channel EEG signal. This signal is then filtered through a 60-Hz notch filter and presented to the *evoked potential averaging computer.* The raw or ongoing EEG signal is available at the 60-Hz filter output. The computer can be set, for example, to give 100 trigger pulses to the visual or auditory stimulator. Each brain response is added synchronously with each trigger in the computer. The computer output is then the *summed response potentials* divided by the number of responses (an *average* of 100 signals). This information can be on magnetic tape or hard-copied onto a strip-chart recorder. A 5 μV_{p-p} *calibration signal* can be used to test the averaging process accuracy.

Figure 13-17 shows an *auditory (tone burst) evoked potential* from a 22-year-old male. Notice the positive response at 5 ms and the negative peak at 15 ms. Early responses probably indicate lower-level brain reactions and later ones, higher-level (cerebral cortex) reactions. Waveform origin is still

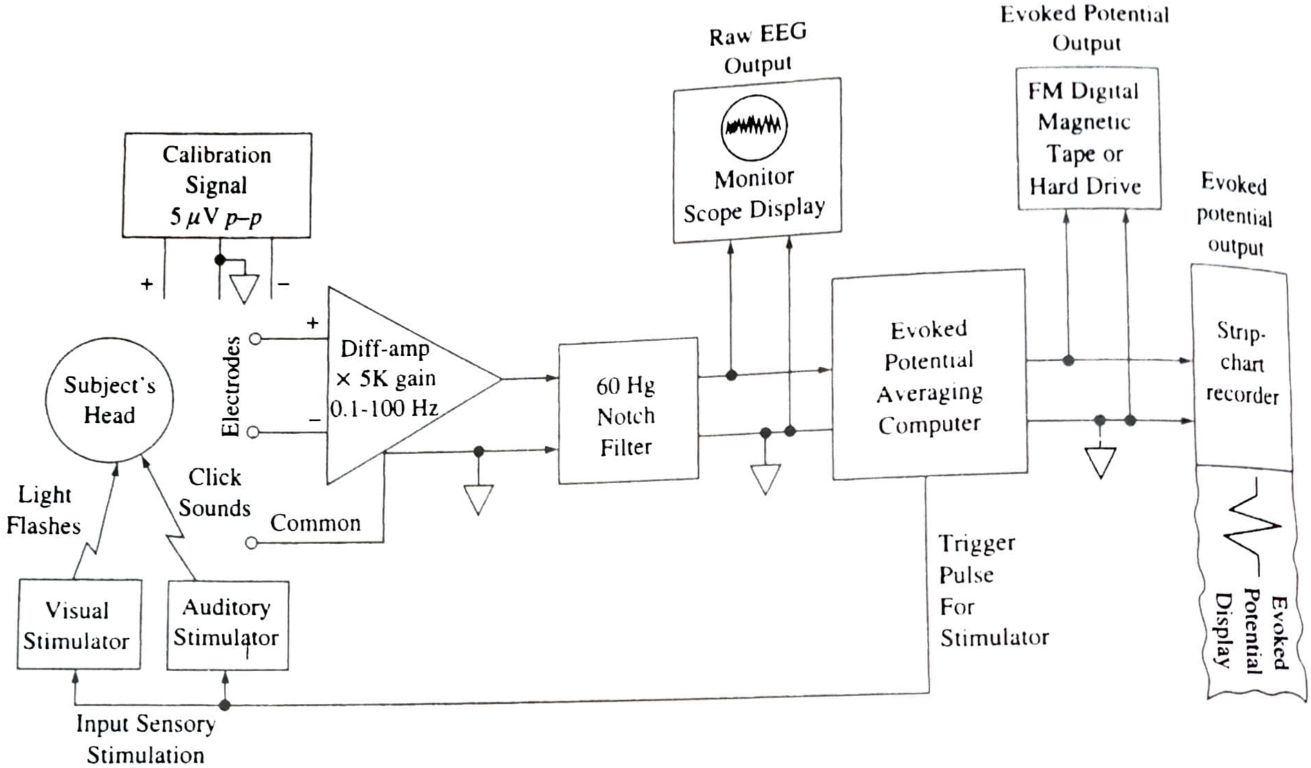


Figure 13-16
Block diagram of a visual and auditory evoked potential system.

being investigated. Evoked potentials can be used to evaluate sensory action and impairment.

13-16 EEG telemetry system

EEG telemetry systems are used to transfer the EEG from the patient's cranium to a remote site without encumbering wires. This technique is useful for children or mentally disturbed persons who may be uncooperative in the data-gathering process. Young children can be monitored while they play; epileptic patients can be monitored while active or just before an attack.

Figure 13-18 shows a block diagram for a two-way telemetry system. From this system, evoked responses can be studied in free-roaming patients. The EEG is amplified and then modulated—amplitude-modulated (AM), frequency-modulated (FM), or pulse-modulated (PM)—and transmitted to a remote site. There, the EEG is demodulated and presented to an EEG machine and evoked-response computer. Tone bursts or clicks can also be transmitted via radio. Radio transmission is, however, plagued with noise interference. Control of the frequency band and power output is required by the Federal Communications Commission (FCC) in the United States. Furthermore, it is difficult to transmit multichannel because bandwidth problems become important. Transmission media other than radio may prove more suitable for line-of-sight EEG telemetry.

13-17 Typical EEG system artifacts, faults, troubleshooting, and maintenance

EEG recording systems suffer from artifacts that can obscure the signals of interest and render diagnosis impossible. Figure 13-19 shows typical artifacts. Aside from 60-Hz interference and *eye blinks* that result in spikes, *muscle activity* from the scalp causes significant interference (Figure 13-19*a*). This can be reduced by adjusting external EEG brainstem. Partial epilepsy almost always results from some organic lesion of the brain, such as a scar that pulls on the neuronal tissue, a tumor that compresses an area of the brain, or a destroyed region of the brain tissue. Lesions such as these can cause local neurons to fire very rapid discharges. When the rate exceeds approximately 1000/s, synchronous waves begin spreading over adjacent cortical regions. These waves presumably result from the activity of localized reverberating neuronal circuits that gradually recruit adjacent areas of the cortex into the "discharge," or firing, zone. The process spreads to adjacent areas at rates as slow as a few millimeters per minute to as fast as several centimeters per minute. When such a wave of excitation spreads over the motor cortex, it causes a progressive "march" of muscular contractions throughout the opposite side of the body, beginning perhaps in the leg region and marching progressively upward to the head region, or at other times marching in the opposite direction. This is called *Jacksonian epilepsy* or *Jacksonian march*.

Another type of partial epilepsy is the so-called *psychomotor seizure*, which may cause (1) a short period of amnesia, (2) an attack of abnormal rage, (3) sudden anxiety or fear, (4) a moment of incoherent speech or mumbling, or (5) a motor act of rubbing the face with the hand, attacking someone, and so forth. Sometimes the person does not remember his or her activities during the attack; at other times the person is completely aware of, but unable to control, his or her behavior. The bottom tracing of Figure 4.27(c) represents a typical EEG during a psychomotor seizure showing a low-frequency rectangular-wave response with a frequency between 2 and 4 Hz with superimposed 14 Hz waves.

The EEG frequently can be used to locate tumors and also abnormal spiking waves originating in diseased brain tissue that might predispose to epileptic attacks. Once such a focal point is found, surgical excision of the focus often prevents future epileptic seizures.

The EEG is also used to monitor the depth of anesthesia.

The EEG is also used as a brain–computer interface to enable disabled persons to communicate with a computer.

4.9 THE MAGNETOENCEPHALOGRAM

Active bioelectric sources in the brain generate magnetic as well as electric fields. However, the magnitude of the magnetic field associated with active cortex is extremely low. For example, it is estimated that the magnetic field of the alpha wave is approximately 0.1 pT at a distance of 5 cm from the surface of the scalp. By way of comparison, this biomagnetic field associated with the magnetoencephalogram (MEG) is roughly one hundred million times weaker than the magnetic field of the earth ($\sim 50\ \mu$T). Recent technological advances in the study of superconductivity have made measurement of these extremely low-strength magnetic fields possible. Specifically, the superconducting quantum interference device (SQUID) magnetometer, which is based on a superconducting effect at liquid helium temperature, has sensitivity on the order of 0.01 pT. Background fields such as the earth's magnetic field and urban magnetic noise fields ($\sim$10 to 100 nT) can be removed for all practical purposes by using a gradiometer technique.

Using the MEG offers a number of advantages: (1) The brain and overlying tissues can be characterized as a single medium having a constant magnetic permeability μ. Therefore, the magnetic field (unlike the electric field) is not influenced by the shell-like anisotropic inhomogeneities (meninges, fluid layers, skull, muscle layer, and scalp) surrounding the brain. (2) The measurement is indirect in that electrodes are not necessary to record the MEG. That is, the SQUID detector does not need to touch the scalp, because the magnetic field does not disappear in air.

The magnetic vector potential **A** has the same orientation as the equivalent current dipole representing an active region of the brain. For a derivation of the vector potential in terms of the volume current density **J**, see Plonsey (1969). Because the magnetic field lies perpendicular to the vector potential, radially oriented current dipoles produce magnetic fields that are oriented tangentially to the sphere representing the head. Similarly, tangentially oriented brain dipoles produce radially oriented magnetic fields.

The local time dependence of biomagnetic fields can be recorded faithfully with SQUID detection systems, but in order to measure the field distribution over the surface of the scalp, measurements must be made at many locations. This is a time-consuming process. Superconducting quantum interference device (SQUID) magnetometer vendors have systems with well over 100 channels (Wiksow, 1995). Advances in material fabrication techniques in the field of superconductivity should yield smaller detector coils for better spatial resolution and, subsequently, more precise localization of intracerebral sources of activity.

PROBLEMS

4.1 What are the four main factors involved in the movement of ions across the cell membrane in the steady-state condition?

4.2 Assume that life on Mars requires an interior cell potential of +100 mV and that the extracellular concentrations of the three major species are as given in Example 4.1. Choose *one* species that has the permeability coefficient given, and assume the other two permeabilities are zero. Design the cell by calculating the intracellular concentration of the chosen species.

4.3 An excitable cell is impaled by a micropipette, and a second extracellular electrode is placed close by at the outer-membrane surface. Brief pulses of current are then passed between these electrodes, which may cause it to conduct an action potential. Explain how the polarity of the stimulating pair influences the membrane potential, and subsequently the activity, of the excitable cell.

of a permanently implanted spinal electrode, with which the paraplegic is able to empty the bladder completely without the use of a catheter.

The stimulating device is similar to that used in phrenic nerve stimulation. However, the stimulus provided is in the form of a biphasic pulse with a pre-set pulse width of 0.2 ms, a pulse intensity of 0.5 to 25 V and a pulse rate of 10 to 50 Hz. The sacral cord electrode consists of two insulated platinum wires, 2.5 mm in length, with 1.5 mm bared conical tips, mounted 2.5 mm apart on an epoxy strip. The two lead out wires are flexible, silastic-coated and are made of stainless steel and connect to a receiver with a circumference of 3 cm. The receiver is placed in the subcutaneous tissue on the left or right side of the patient's waist. Voiding usually begins 10 to 15 seconds after the onset of stimulation.

29.9 DEEP BRAIN STIMULATION (DBS)

The deep brain stimulation (DBS) therapy is a new treatment technique for a number of neurologic disorders. The technique in select brain regions has provided remarkable therapeutic benefits for otherwise treatment-resistant movement disorders such as Parkinson's disease, tremor and dystonia.

The system consists of three components: the implanted pulse generator (neurostimulator), the electrode and the extension. The arrangement is shown in Fig. 29.19. The electrode or lead is a thin, insulated wire which is inserted through a small opening in the skull and implanted in the brain. The tip of the electrode is positioned within the targeted brain area. The extension is an insulated wire that is passed under the skin of the head, neck, and shoulder, connecting the lead to the neurostimulator which is usually implanted under the skin near the collarbone. The stimulator delivers a constant fast-frequency stimulus which interrupts a specific circuit in the brain that is overactive in the disease state. This interruption of the diseased overactive circuit can significantly improve the symptoms of the disease.

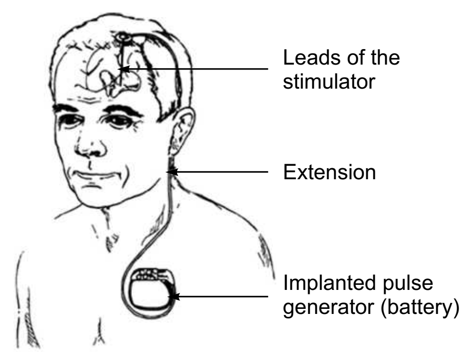


Fig. 29.19 *Arrangement for deep brain stimulation for Parkinson's disease*

Fig. 29.20 is a typical neurostimulator from M/s Medtronic which is a dual channel device capable of delivering bilateral stimulation. It has two modes of operation. In the voltage mode, it delivers pulses with a rate of 2 to 250 Hz and voltage level of 0 to 10.5 volts. In the current mode, the pulse frequency is 30 to 250 Hz and current range of 0 to 25.5 mA. In both the modes, the pulse width is kept between 60 to 450 μs. Each lead carries upto 4 electrodes. The device contains a non-rechargeable battery and microelectronic circuitry to deliver a controlled electrical pulse to precisely targeted areas of the brain. The device is typically implanted subcutaneously near the clavicle.

Fig. 29.20 *Neuro stimulator (Courtesy: M/s Medtronic)*

CHAPTER

11

Biomedical Telemetry

11.1 BIOTELEMETRY

Biotelemetry is the use of telemetry methods in order to remotely observe, document, and measure certain physiological functions in human beings or other living organisms. The field consists of several subfields, including medical and human research telemetry, animal telemetry, and implantable biotelemetry. Medical telemetry is of particular importance because it can be used to remotely track the vital signs of ambulatory patients. Generally, a biotelemetry system used for this purpose measures functions like body temperature, heart rate, blood pressure, and muscle movement.

The use of biotelemetry systems began as early as the late 1950s, during the space race era. At that time, these systems were used to record physiological signs from animals or humans who traveled to outer space in a space vehicle. The signals were then transmitted back to a space station on earth for observation and study.

Measurements which have been done in biotelemetry can be determined in two categories:

- Bioelectrical variables, such as electrocardiogram (ECG), electromyogram (EMG) and electroencephalogram (EEG).
- Physiological variables that require transducers, such as blood pressure, gastrointestinal pressure, blood flow and temperature. By using suitable transducers, telemetry can be employed for the measurement of a wide variety of physiological variables.

11.1.1 Wireless Telemetry

Wireless telemetry permits examination of the physiological data of man or animal under normal conditions and in natural surroundings without any discomfort or obstruction to the person or animal under investigation. Factors influencing healthy and sick persons during the performance of their daily tasks may thus be easily recognized and evaluated. Wireless bio-telemetry has made possible the study of active subjects under conditions that so far prohibited measurements. It is, therefore, an indispensable technique in situations where no cable connection is feasible (Gandikola, 2000).

Using wireless telemetry, physiological signals can be obtained from swimmers, riders, athletes, pilots or manual labourers. Telemetric surveillance is most convenient during transportation within the hospital area as well for the continuous monitoring of patients sent to other wards or clinics for check-up or therapy.

Most biotelemetry systems are wireless. Usually, they consist of several components, including sensors, transmitters, a radio antenna, and a receiver. Patients or animal subjects typically wear the transmitters on the outside of their bodies. Signals are then sent from the transmitters to a receiver in the biotelemetry lab to be reviewed and analyzed. A display unit in the lab allows the staff to see vital sign information from several different patients or animals at one time.

In particular, cardiovascular patients benefit from the use of wireless biotelemetric systems. These devices offer cardiac patients the ability to stay mobile while being observed. The systems used for these patients usually depend on radio-frequency communications to monitor heart rates, blood flow, and blood pressure. This is all done without requiring the patient to be hooked up to a bedside monitor with a wired connection.

Biotelemetry can also be used to conduct research on animal behaviour in their natural environments or on animal migration patterns. Typically, this research is conducted by placing a transmitter on the animal. Biologists then track the animal by following the transmittal signal. Even on sleeping mammals or birds, animal telemetry devices usually record everything from respiration, heart rate, and heart muscle activity to neural and cardiac movements.

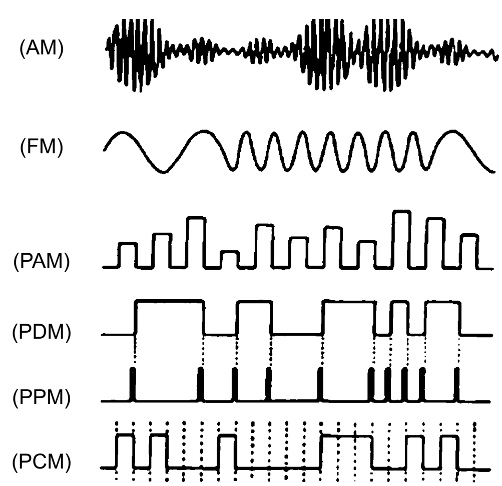
When implants are used in biotelemetry, it usually means that the transmitter devices are implanted in the animal or human being studied. For example, cochlear implants usually have built-in telemetry systems that allow the internal device to be monitored. More powerful transmitters can be more difficult to implant in a subject, and strong transmitters with large batteries can impact a subject's behaviour or energy levels.

11.1.2 Modulation Systems

The physiological signal alone cannot be transmitted with reasonable efficiency as a radio wave because of its low frequency. Therefore a high-frequency carrier wave is usually used as a transmitting vehicle. The signal information is forced upon the carrier by modulation, that is, by a signal-dependent (and thus time-variant) characteristic parameter such as amplitude, frequency, or phase (Stremler, 1982).

Amplitude modulation (AM) is a simple method of signal transmission; its application, however, is rather limited. AM signals are more adversely affected by sources of interference that additively superimpose the carrier than are the signals modulated by other procedures. Frequency modulation (FM) provides absolute values for the signal amplitudes and noise interferes much less with the reception than in AM systems.

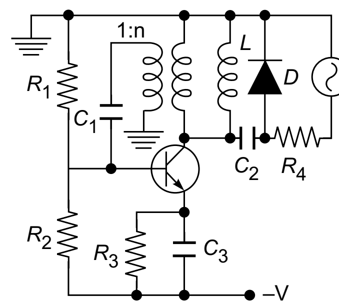
In pulse-modulation techniques, only discrete samples representing the signal are used to modulate the carrier. In pulse modulation the discrete samples are used to vary a parameter of a pulse waveform. Such parameters are the amplitude (pulse amplitude modulation, PAM), duration (pulse duration modulation—PDM), or position (pulse position modulation—PPM) of the pulse. Representative waveforms for the different modulation techniques are shown in Fig. 11.1. Another coding system is pulse code modulation (PCM), in which the analog signal is converted to digital sample points and the series of binary digits is transmitted.



➤ Fig. 11.1 *Modulation and coding techniques*

The modulation systems used in wireless telemetry for transmitting biomedical signals makes use of two modulators. This means that a comparatively lower frequency sub-carrier is employed in addition to the VHF (very high frequency), which finally transmits the signal from the transmitter. The principle of double modulation gives better interference free performance in transmission and enables the reception of low frequency biological signals. The sub-modulator can be a FM (frequency modulation) system or a PWM (Pulse Width Modulation) system, whereas the final modulator is practically always an FM system.

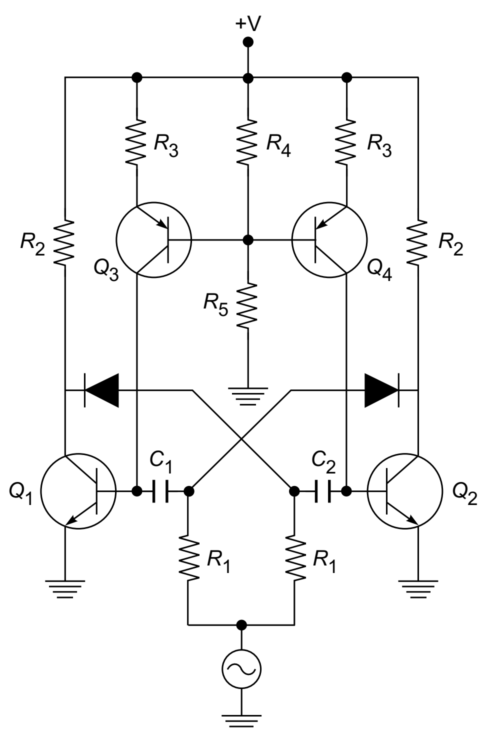
Frequency Modulation: In frequency modulation, intelligence is transmitted by varying the instantaneous frequency in accordance with the signal to be modulated on the wave, while keeping the amplitude of the carrier wave constant. The rate at which the instantaneous frequency varies is the modulating frequency. The magnitude to which the carrier frequency varies away from the centre frequency is called "Frequency Deviation" and is proportional to the amplitude of the modulating signal. Usually, an FM signal is produced by controlling the frequency of an oscillator by the amplitude of the modulating voltage. For example, the frequency of oscillation for most oscillators depends on a particular value of capacitance. If the modulation signal can be applied in such a way that it changes the value of capacitance, then the frequency of oscillation will change in accordance with the amplitude of the modulating signal.



➤ **Fig. 11.2** ***Circuit diagram of a frequency modulator using varactor diode***

Fig. 11.2 shows a tuned oscillator that serves as a frequency modulator. The diode used is a varactor diode operating in the reverse-biased mode and, therefore, presents a depletion layer capacitance to the tank circuit. This capacitance is a function of the reverse-biased voltage across the diode and, therefore, produces an FM wave with the modulating signal applied as shown in the circuit diagram. This type of circuit can allow frequency deviations of 2–5% of the carrier frequency without serious distortion.

Pulse Width Modulation: Pulse width modulation method has the advantage of being less perceptive to distortion and noise. Fig. 11.3 shows a typical pulse width modulator. Transistors Q_1 and Q_2 form a free-running multi-vibrator. Transistors Q_3 and Q_4 provide constant current sources for charging the timing capacitors C_1 and C_2 and driving transistors Q_1 and Q_2. When Q_1 is 'off' and Q_2 is 'on', capacitor C_2 charges through R_1 to the amplitude of the modulating voltage e_m. The other side of this capacitor is connected to the base of transistor Q_2 and



➤ **Fig. 11.3** ***Pulse width modulator*** is at zero volt. When Q_1 turns 'on' switching the circuit to the other stage, the base voltage of Q_2 drops from approximately zero to $-e_m$. Transistor Q_2 will remain 'off' until the base voltage charges to zero volt. Since the charging current is constant at I, the time required to charge C_2 and restore the circuit to the initial stage is:

$$T_2 = \frac{C_2}{I} . e_m$$

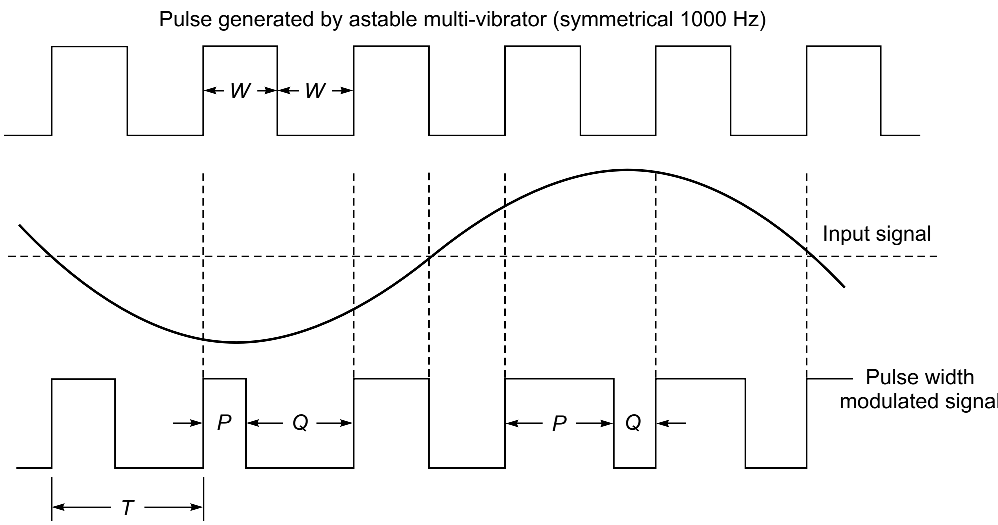
Similarly, the time that the circuit remains in the original stage is:

$$T_1 = \frac{C_1}{I} . e_m$$

This shows that both portions of the astable period are directly proportional to the modulating voltage.

When a balanced differential output from an amplifier such as the ECG amplifier is applied to the input points 1 and 2, the frequency of the astable multi-vibrator would remain constant, but the width of the pulse available at the collector of transistor Q_2 shall vary in accordance with the amplitude of the input signal.

In practice, the negative edge of the square wave is varied in rhythm with the ECG signal. Therefore, only this edge contains information of interest. The ratio P:Q (Fig. 11.4) represents the momentary amplitude of the ECG. The amplitude or even the frequency variation of the square wave does not have an influence on the P:Q ratio and consequently on the ECG signal. The signal output from this modulator is fed to a normal speech transmitter, usually via an attenuator, to make it suitable to the input level of the transmitter.



➤ **Fig. 11.4** ***Variation of pulse width with amplitude of the input signal***

11.1.3 Choice of Radio Carrier Frequency

In every country there are regulations governing the use of only certain frequency and bandwidth for medical telemetry. Therefore, the permission to operate a particular telemetry system needs to be obtained from the concerned department of the country concerned. The radio frequencies normally used for medical telemetry purposes are of the order of 37, 102, 153, 159, 220 and 450 MHz. The transmitter is typically of 50 mW at 50 Ω, which can give a transmission range of about 1.5 km in the open flat country. The range will be much less in built-up areas. In USA, two frequency bands have been designated for short range medical telemetry work by the FCC (Federal Communications Commission). The lower frequency band of 174–216 MHz, coincides with the VHF television broadcast band (Channels 7–13). Therefore, the output of the telemetry transmitter must be limited to avoid interference with TV sets. Operation of telemetry units in this band does not normally require any licence. In the higher frequency band of 450–470 MHz, greater transmitter power is allowed, but an FCC licence has to be obtained for operating the system.

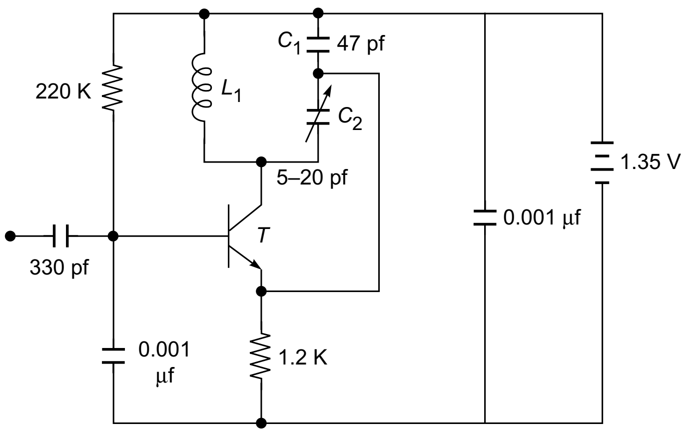
Radio waves can travel through most non-conducting material such as air, wood, and plaster with relative ease. However, they are hindered, blocked or reflected by most conductive material and by concrete because of the presence of reinforced steel. Therefore, transmission may be lost or be of poor quality when a patient with a telemetry transmitter moves in an environment with a concrete wall or behind a structural column. Reception may also get affected by radio frequency wave effects that may result in areas of poor reception or null spots, under some conditions of patient location and carrier frequency. Another serious problem that is sometimes present in the telemetry systems is the cross-talk or interference between telemetry channels. It can be minimized by the careful selection of transmitter frequencies, by the use of a suitable antenna system and by the equipment design.

High transmission frequencies are chosen for radio-wave transmitters for several reasons. At high frequencies, passive components such as capacitors and coils have very small size to facilitate miniaturization, high data transfer capacity is enabled, the small wavelengths do not need large antennas, and in-house transmission is possible over longer distances in spite of metallic constructions.

The range of any radio system is primarily determined by transmitter output power and frequency. The use of a higher-powered transmitter than is required for adequate range is preferable as it may eliminate or reduce some noise effects due to interference from other sources.

11.1.4 Transmitter

Fig. 11.5 shows a circuit diagram of the FM transmitter stage commonly used in medical telemetry. The transistor T acts in a grounded base Colpitts R.F oscillator with L_1 and C_1 and C_2 as the tank circuit. The positive feedback to the emitter is provided from a capacitive divider in the collector circuit formed by C_1 and C_2. Inductor L_1 functions both as a tuning coil and a transmitting antenna. Trim capacitor C_2 is adjusted to precisely set the transmission frequency at the desired point. In this case, it is within the standard FM broadcast band from 88 to 108 MHz. Frequency modulation is achieved by variation in the operating point of the transistor, which in turn varies its collector capacitance, thus changing the resonant frequency of the tank circuit. The operating point is changed by the sub-carrier input. Thus, the transmitter's output consists of an RF signal, tuned in the FM broadcast band and frequency modulated by the sub-carrier oscillator (SCO), which in turn is frequency modulated by the physiological signals of interest. It is better to use a separate power source for the RF oscillator from other parts of the circuit to achieve stability and prevent interference between circuit functions (Beerwinkle and Burch, 1976).



➤ Fig. 11.5 *Typical circuit diagram of an FM telemetry transmitter*

11.1.5 The Receiver

In most cases, the receiver can be a common broadcast receiver with a sensitivity of 1µV. The output of the HF unit of the receiver is fed to the sub-demodulator to extract the modulating signal. In a FM/FM system, the sub-demodulator first converts the FM signal into an AM signal. This is followed by an AM detector which demodulates the newly created AM waveform. With this arrangement, the output is linear with frequency deviation only for small frequency deviations. Other types of detectors can be used to improve the linearity.

In the PWM/FM system, a square wave is obtained at the output of the RF unit. This square wave is clipped to cut off all amplitude variations of the incoming square wave and the average value of the normalized square wave is determined. The value thus obtained is directly proportional to the area which in turn is directly proportional to the pulse duration. Since the pulse duration is directly proportional to the modulating frequency, the output signal is directly proportional to the output voltage of the demodulator. The output voltage of the demodulator is adjusted such that it can be directly fed to a chart recorder. The receiver unit also provides signal outputs to store the demodulated signal in an external memory.

Successful utilization of biological telemetry systems is usually dependent upon the user's systematic understanding of the limits of the system, both biological and electrical. The two major areas of difficulty arising in biotelemetry occur at the system interfaces. The first is the interface between the biological system and the electrical system. No amount of engineering can correct a shoddy, hasty job of instrumenting the subject. Therefore, electrodes and transducers must be put on with great care. The other major area of difficulty is the interface between the transmitter and receiver. It must be kept in mind that the range of operation should be limited to the primary service area, otherwise the fringe area reception is likely to be noisy and unacceptable. Besides this, there are problems caused by the patient's movements, by widely varying signal strength and because of interference from electrical equipment and other radio systems, which need careful equipment design and operating procedures.

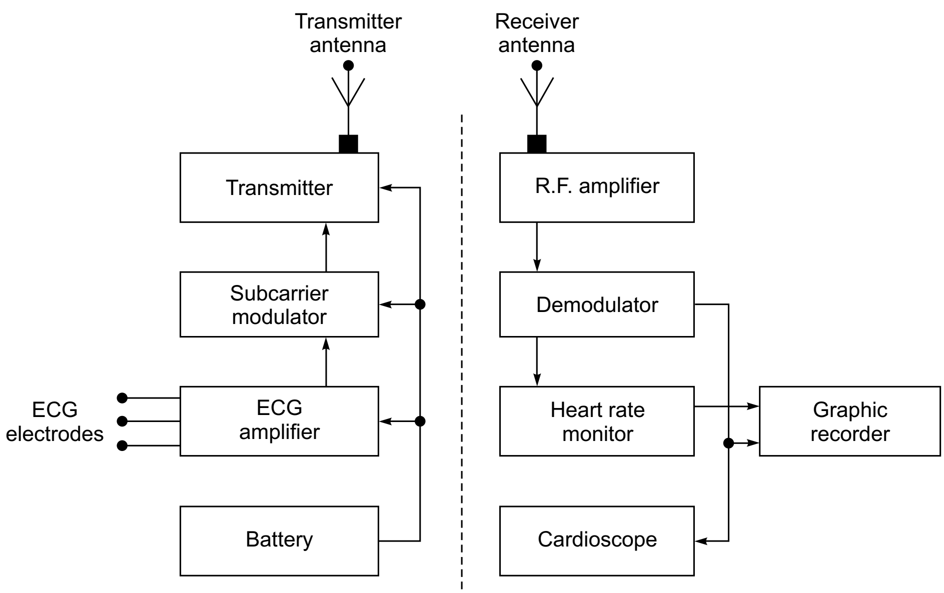
11.2 SINGLE CHANNEL TELEMETRY SYSTEMS

In a majority of the situations requiring monitoring of the patients by wireless telemetry, the parameter which is most commonly studied is the electrocardiogram. It is known that the display of the ECG and cardiac rate gives sufficient information on the loading of the cardiovascular system of the active subjects. Therefore, we shall first deal with a single channel telemetry system suitable for the transmission of an electrocardiogram.

11.2.1 ECG Telemetry System

Fig. 11.6 shows the block diagram of a single channel telemetry system suitable for the transmission of an electrocardiogram. There are two main parts:

- The Telemetry Transmitter which consists of an ECG amplifier, a sub-carrier oscillator and a UHF transmitter along with dry cell batteries.
- Telemetry Receiver consists of a high frequency unit and a demodulator, to which an electrocardiograph can be connected to record, a cardioscope to display and a memory device to store the ECG. A heart rate meter with an alarm facility can be provided to continuously monitor the beat-to-beat heart rate of the subject.



➤ Fig. 11.6 *Block diagram of a single channel telemetry system*

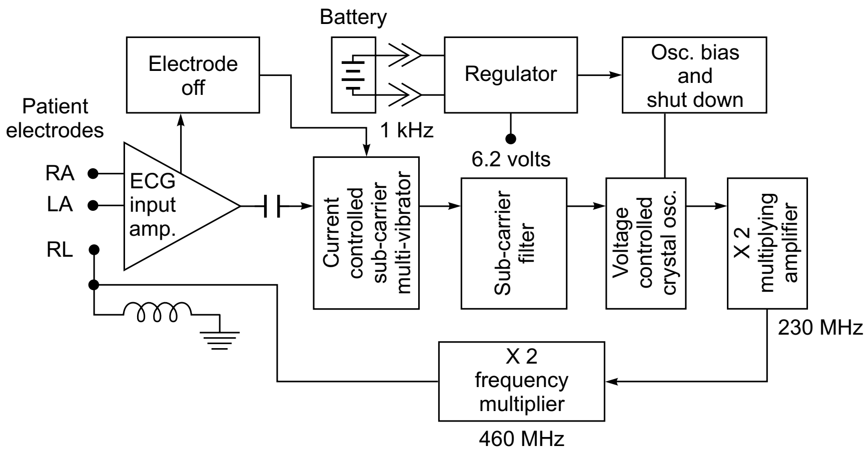
For distortion-free transmission of ECG, the following requirements must be met (Kuiper *et al.*, 1996):

- The subject should be able to carry on with his normal activities whilst carrying the instruments without the slightest discomfort. He should be able to forget their presence after some minutes of application.
- Motion artefacts and muscle potential interference should be kept minimum.
- The battery life should be long enough so that a complete experimental procedure may be carried out.

- While monitoring paced patients for ECG through telemetry, it is necessary to reduce pacemaker pulses. The amplitude of pacemaker pulses can be as large as 80 mV compared to 1–2 mV, which is typical of the ECG. The ECG amplifiers in the transmitter are slew rate (rate of change of output) limited so that the relatively narrow pacemaker pulses are reduced in amplitude substantially.

Some ECG telemetry systems operate in the 450–470 MHz band, which is well-suited for transmission within a hospital and has the added advantage of having a large number of channels available. The circuit details of an ECG telemetry system are described below:

Transmitter: A block diagram of the transmitter is shown in Fig. 11.7. The ECG signal, picked up by three pre-gelled electrodes attached to the patient's chest, is amplified and used to frequency modulate a 1 kHz sub-carrier that in turn frequency-modulates the UHF carrier. The resulting signal is radiated by one of the electrode leads (RL), which serves as the antenna. The input circuitry is protected against large amplitude pulses that may result during defibrillation.



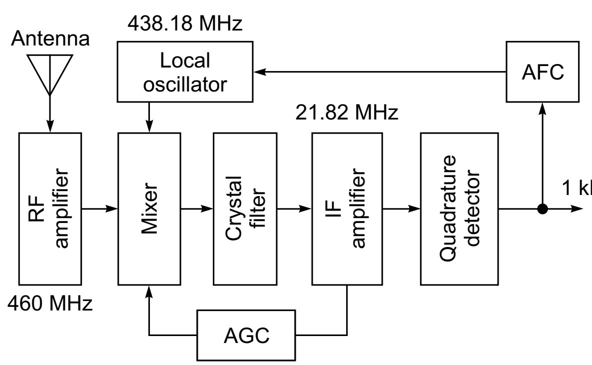
➤ **Fig. 11.7** ***Block diagram of ECG telemetry transmitter (Redrawn after Larsen et al., 1972; by permission of Hewlett Packard, U.S.A.)***

The ECG input amplifier is ac coupled to the succeeding stages. The coupling capacitor not only eliminates dc voltage that results from the contact potentials at the patient-electrode interface, but also determines the low-frequency cut-off of the system which is usually 0.4 Hz. The sub-carrier oscillator is a current-controlled multi-vibrator which provides ±320 Hz deviation from the 1 kHz centre frequency for a full range (± 5 mV) ECG signal. The sub-carrier filter removes the square-wave harmonic and results in a sinusoid for modulating the RF carrier. In the event of one of the electrodes failing off, the frequency of the multi-vibrator shifts by about 400 Hz. This condition when sensed in the receiver turns on an 'Electrode inoperative' alarm.

The carrier is generated in a crystal-controlled oscillator operating at 115 MHz. The crystal is a fifth overtone device and is connected and operated in the series resonant mode. This is followed by two frequency doubler stages. The first stage is a class-C transistor doubler and the second is a series connected step recovery diode doubler. With the output power around 2 mW, the system has an operating range of 60 m within a hospital.

***Receiver*:** The receiver uses an omnidirectional receiving antenna which is a quarter-wave monopole, mounted vertically over the ground plane of the receiver top cover. This arrangement works well to pick up the randomly polarized signals transmitted by moving patients.

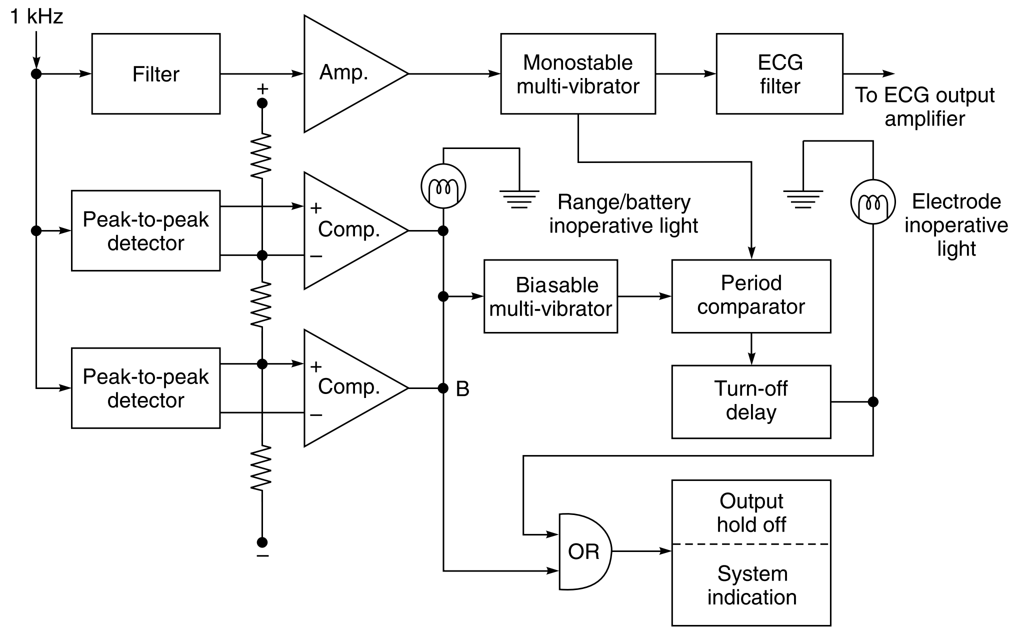
The receiver (Fig. 11.8) comprises an RF amplifier, which provides a low noise figure, RF filtering and image-frequency rejection. In addition to this, the RF amplifier also suppresses local oscillator radiation to –60 dBm to minimize the possibility of cross-coupling where several receivers are used in one central station. The local oscillator employs a crystal (115 MHz) similar to the one in the transmitter and x4 multiplier and a tuned amplifier. The mixer uses the square law characteristics of a FET to avoid interference problems due to third-order intermodulation. The mixer is followed by an 8-pole crystal filter that determines the receiver selectivity. This filter with a 10 kHz bandwidth provides 60 dB of rejection for signals 13 kHz from the IF centre frequency (21.82 MHz). The IF amplifier provides the requisite gain stages and operates an AGC amplifier which reduces the mixer gain under strong signal conditions to avoid overloading at the IF stages. The IF amplifier is followed by a discriminator, a quadrature detector. The output of the discriminator is the 1 kHz sub-carrier. This output is averaged and fed back to the local oscillator for automatic frequency control. The 1 kHz sub-carrier is demodulated to convert frequency-to-voltage to recover the original ECG waveform. The ECG is passed through a low-pass filter (Fig. 11.9) having a cut-off frequency of 50 Hz and then given to a monitoring instrument. The 1 kHz sub-carrier is examined to determine whether or not a satisfactory signal is being received. This is done by establishing a window of acceptability for the sub-carrier amplitude. If the amplitude is within the window, then the received signal is considered valid. In the case of AM or FM interference, an 'inoperative' alarm lamp lights up.



➤ Fig. 11.8 ***Block diagram of high frequency section of ECG telemetry receiver (Adapted from Larsen et al., 1972; by permission Hewlett Packard, U.S.A.)***

Different manufacturers use different carrier frequencies in their telemetry equipment. Use of the FM television band covering 174 to 185.7 MHz (VHF TV channels 7 and 8) is quite common. However, the output is limited to a maximum of 150 µV/m at a distance of 30 m to eliminate interference with commercial television channels.

Some transmitters are also provided with special arrangements like low transmitter battery and nurse call facility. In both these situations, a fixed frequency signal is generated, which causes a deviation of the sub-carrier and when received at the receiver, actuate appropriate circuitry for visual indications.



➤ Fig. 11.9 *Schematic diagram of ECG demodulation and 'inoperate' circuits in ECG telemetry receiver (After Larsen et al., 1972; by permission of Hewlett Packard, U.S.A.)*

Also, some telemetry systems include an out-of-range indication facility. This condition is caused by a patient lying on the leads or the patient getting out of range from the receiver capability. For this, the RF carrier signal from the transmitter is continuously monitored. When this signal level falls below the limit set, the alarm will turn on.

For the satisfactory operation of a radiotelemetry system, it is important to have proper orientation between the transmitting and receiving antennas. There can be orientations in which none of the signals radiated from the transmitting antenna are picked up by the receiving antenna. It is therefore important to have some means for indicating when signal interference or signal dropout is occurring. Such a signal makes it possible to take steps to rectify this problem and informs the clinical staff that the information being received is noise and should be disregarded.

11.2.2 Temperature Telemetry System

Systems for the transmission of alternating potentials representing such parameters as ECG, EEG and EMG are relatively easy to construct. Telemetry systems which are sufficiently stable to telemeter direct current outputs from temperature, pressure or other similar transducers continuously for long periods present greater design problems. In such cases, the information is conveyed as a modulation of the mark/space ratio of a square wave. A temperature telemetry system based on this principle is illustrated in the circuit shown in Fig. 11.10.

Temperature is sensed by a thermistor having a resistance of 100 Ω (at 20°C) placed in the emitter of transistor T_1. Transistors T_1 and T_2 form a multi-vibrator circuit timed by the thermistor, $R_1 + R_2$ and C_1. R_1 is adjusted to give 1:1 mark/space ratio at midscale temperature (35–41°C). The multi-vibrator produces a square wave output at about 200 Hz. Its frequency is